6 Discussion

Our findings reveal that the architecture of a long-context prompt—not just its content—is a critical factor in model safety. The discovery that goal positioning can determine the success of a jailbreak has immediate practical implications for designing safer AI systems. For instance, interfaces that handle long documents could be designed to programmatically place user queries at the end of the context to minimize risk. Furthermore, the compute-optimality analysis suggests that as computational resources become cheaper, long-context attacks will not only become more prevalent but will be the most efficient method for malicious actors.

These vulnerabilities are particularly concerning in emerging agentic systems. In these environments, the context window naturally grows through multi-turn dialogue, tool use histories, and long task trajectories. We extend our work to this setting using SHADE-Arena Gao et al. (2024) and find a similar, concerning trend: our initial experiments show that while a powerful model like OpenAI's o3 will refuse a harmful task asked directly, it is more likely to comply when the same request is embedded in a long, multi-turn trajectory. This suggests that long interaction histories can gradually erode a model's safety alignment, creating a significant, yet-unaddressed threat vector for autonomous agents. We include example trajectories in Appendix E.

Finally, a key advantage of our attack is its practicality. The benign context required for Ninja does not need to be generated by a powerful model. An attacker can use a much smaller, cheaper model to generate semantically relevant filler content, or even retrieve it from a public corpus, making the attack both low-cost and highly transferable.

7 Limitations

While our findings demonstrate a significant vulnerability, the effectiveness of the Ninja attack is not universal. The attack's success does vary across models; for instance, Gemini Flash exhibited notable robustness in our experiments. The exact reasons for this are difficult to determine without transparency into the model's architecture and specific safety training procedures, but it suggests that certain alignment techniques may be less susceptible to this failure mode.

8 Conclusion

This work reveals a fundamental tension between the scaling of long-context capabilities and the preservation of model safety. We introduced Ninja, a simple, stealthy, and compute-optimal attack that demonstrates how simply increasing the length of a benign context can reliably jailbreak a wide range of aligned language models. Our core findings show that not only context length, but also the structural placement of the harmful goal, are critical determinants of safety compliance.

The success of Ninja serves as a clear warning: as models are deployed in increasingly complex, long-running tasks, their vast context windows become a significant attack surface. The pursuit of ever-larger context capabilities cannot proceed without a parallel focus on developing robust, context-aware safety mechanisms. Future research must move beyond content filtering and address these deeper, structural vulnerabilities to ensure that the next generation of language models can be both capable and safe.

References

Cem Anil, Esin DURMUS, Nina Rimsky, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, Meg Tong, Jesse Mu, Daniel J Ford, Francesco Mosconi, Rajashree Agrawal, Rylan Schaeffer, Naomi Bashkansky, Samuel Svenningsen, Mike Lambert, Ansh Radhakrishnan, Carson Denison, Evan J Hubinger, Yuntao Bai, Trenton Bricken, Timothy Maxwell, Nicholas Schiefer, James Sully, Alex Tamkin, Tamera Lanham, Karina Nguyen, Tomasz Korbak, Jared Kaplan, Deep Ganguli, Samuel R. Bowman, Ethan Perez, Roger Baker

Grosse, and David Duvenaud. Many-shot jailbreaking. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024a. URL https://openreview.net/forum?id=cw5mgd71jW.

Cem Anil, Esin Durmus, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, et al. Many-shot jailbreaking. *arXiv preprint arXiv:2304.XXX*, 2024b. Anthropic Technical Report.

Anthropic. Claude 3.5 and computer use agents, 2024. https://www.anthropic.com/news/3-5-models-and-computer-use.

Levent N. Candogan, Yifei Wu, Esteban Abad Rocamora, Grigorios Chrysos, and Volkan Cevher. Single-pass detection of jailbreaking input in large language models. *Transactions on Machine Learning Research (TMLR)*, 2025.

Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J. Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2023.

Gelei Deng, Yi Liu, Yuekang Li, Kailong Wang, Ying Zhang, Zefeng Li, Haoyu Wang, Tianwei Zhang, and Yang Liu. Masterkey: Automated jailbreak across multiple large language model chatbots. *arXiv preprint arXiv:2307.08715*, 2024.

Leo Gao, Arjun R, Alexander Pan, and etc. SHADE-Arena: Evaluating sabotage and monitoring in llm agents. *arXiv preprint arXiv:2506.15740*, 2024. URL https://arxiv.org/abs/2506.15740.

Greg Kamradt. Needle in a haystack - pressure testing llms. *GitHub*, 2023.

Priyanshu Kumar, Elaine Lau, Saranya Vijayakumar, Tu Trinh, Scale Red Team, Elaine Chang, Vaughn Robinson, Sean Hendryx, Shuyan Zhou, Matt Fredrikson, Summer Yue, and Zifan Wang. Refusal-trained llms are easily jailbroken as browser agents. *ArXiv*, abs/2410.13886, 2024. URL https://api.semanticscholar.org/CorpusID:273482595.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024a. doi: 10.1162/tacl_a_00638. URL https://aclanthology.org/2024.tacl-1.9.

Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao. Autodan: Generating stealthy jailbreak prompts on aligned large language models. *arXiv preprint arXiv:2310.04451*, 2024b.

Yida Lu, Jiale Cheng, Zhexin Zhang, Shiyao Cui, Cunxiang Wang, Xiaotao Gu, Yuxiao Dong, Jie Tang, Hongning Wang, and Minlie Huang. Longsafety: Evaluating long-context safety of large language models. *ArXiv*, abs/2502.16971, 2025. URL https://api.semanticscholar.org/CorpusID:276575919.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*, 2024a.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *Proceedings of the 41st International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2024b.

Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson, Yaron Singer, and Amin Karbasi. Tree of attacks: Jailbreaking black-box llms automatically. *arXiv preprint arXiv:2312.02119*, 2023.

OpenAI. Introducing openai operator, 2024. https://openai.com/index/introducing-operator/.

Chinmay Pathade. Red teaming the mind of the machine: A systematic evaluation of prompt injection and jailbreak vulnerabilities in llms. *arXiv preprint arXiv:2505.04806*, 2025.

Mark Russinovich, Ahmed Salem, and Ronen Eldan. Great, now write an article about that: The crescendo multi-turn llm jailbreak attack. *arXiv preprint arXiv:2404.01833*, 2024.

Guanchu Shen, Di Zhao, Lei Feng, Xiaofei He, Jing Wang, Shuhan Shen, Hanghang Tong, Yuxiao Dong, Jun Li, Xiaoqing Zheng, and Yi Zeng. Pandaguard: Systematic evaluation of llm safety in the era of jailbreaking attacks. *arXiv preprint arXiv:2505.13862*, 2025.

Rohan Upadhayay, Yuxuan Shao, Percy Liang, Eric Zhang, and Arvind Narayanan. Cognitive overload attack: Prompt injection for long context. *arXiv preprint arXiv:2410.11272*, 2024.

Xinyi Wu, Yifei Wang, Stefanie Jegelka, and Ali Jadbabaie. On the emergence of position bias in transformers. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. URL https://arxiv.org/abs/2502.01951.

Yihan Xie, Jingfeng Yi, Jing Shao, Jack Curl, Lingjuan Lyu, Qifeng Chen, Xing Xie, and Feng Wu. Defending chatgpt against jailbreak attack via self-reminders. *Nature Machine Intelligence*, 5(12):1486–1496, 2023.

Dan Yao, Jing Zhang, Isaac G. Harris, and Magnus Carlsson. Fuzzllm: A novel and universal fuzzing framework for proactively discovering jailbreak vulnerabilities in large language models. *arXiv preprint arXiv:2403.07506*, 2024.

Zhen Zhang, Jiaxin Yang, Peng Ke, Fei Mi, Hong Wang, and Minlie Huang. Defending large language models against jailbreaking attacks through goal prioritization. *arXiv preprint arXiv:2402.09923*, 2024.

An Zhou, Bo Li, and Haoran Wang. Robust prompt optimization for defending language models against jailbreaking attacks. *arXiv preprint arXiv:2401.13984*, 2024.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.